

Assessing Cognitive Fatigue in Human Teammates During Human-Robot Collaboration

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Abstract

As human-robot teams become increasingly integrated into predominantly human environments, such as workplaces, hospitals, and even our homes, understanding cognitive fatigue on human teammates is crucial for the robots. This chapter explores the challenge of assessing cognitive fatigue in an online manner through sensory data during Human Robot Collaboration (HRC). We examine state-of-the-art approaches that leverage signals such as Electroencephalogram (EEG), Electrocardiography (ECG), Galvanic Skin Response (GSR), speech-based markers, and others to assess Cognitive Fatigue (CF) states in real-time. The chapter discusses the strengths and limitations of each modality, the available datasets and the state-of-the-art algorithms for CF assessment, and the implications for adaptive systems that can respond to signs of fatigue. By bridging insights from robotics, neuroscience, and human factors, this chapter provides a comprehensive foundation for designing robust, responsive robot teammates capable of acting when human teammates are cognitively fatigued.

Keywords: speech, human-robot collaboration (HRC), natural language understanding, collaborative robots, Electroencephalogram (EEG), Electrocardiography (ECG), Galvanic Skin Response (GSR), VAS-F

1 Introduction

As HRC becomes increasingly prevalent across various domains, including industry, healthcare, and domestic environments, understanding CF in human teammates has emerged as a critical concern. Wylie *et al.* [1] defined CF as a state of diminished cognitive resources that can result from prolonged or sustained mental effort. It is characterized by a progressive decline in cognitive efficiency, attention, and decision-making abilities, which can degrade task performance and compromise safety outcomes. While mental effort refers to the active mobilization of cognitive resources, such as working memory capacity, required to perform a task effectively [2], CF develops when these cognitive demands are maintained over extended periods without sufficient recovery, leading to the depletion of those cognitive resources. Distinguishing between mental effort and CF, and understanding interplay is essential in HRC settings where humans are often required to sustain high levels of cognitive demand while interacting with robotic teammates.

Humans collaborating with robots while experiencing diminished or fluctuating CF are more susceptible to slower response times, decision-making errors, and lapses in attention, which degrade team performance and compromise safety outcomes [3]. Beyond these direct performance impacts, CF can also alter interpersonal dynamics within the team. Cognitively fatigued users may exhibit changes in trust and engagement toward their robotic counterparts, undermining the overall quality and effectiveness of the collaboration. For instance, Guo *et al.* [4] examined the effects of Cognitive Workload (CWL) on human trust in hybrid HRC scenarios, revealing that elevated CWL significantly modulates trust in robots and may contribute to increased CF, further influencing users' trust and engagement behaviors. Similarly, Matuz *et al.* [5] emphasized that prolonged exposure to high CWL exacerbates CF, reinforcing the broader recognition that continuous monitoring and effective management of human CF are essential for maintaining safe, efficient, and resilient HRC teams.

Zand *et al.* [6] emphasized that CF can be assessed using a range of methods, with the optimal choice depending on the timing, context, and specific demands of the HRC scenario. They also emphasized that a key distinction exists between offline and online assessment approaches. Offline assessments evaluate CF retrospectively, typically after task completion, using self-reported questionnaires, performance metrics, or physiological data. In contrast, online assessments enable continuous monitoring of CF levels to analyze streaming data from sensors or system logs. This provides dynamic insights into the user's cognitive state as the HRC unfolds, supporting adaptive responses from robotic systems. Building upon online assessment, real-time systems not only monitor streaming data but also process and interpret it instantaneously to support adaptive decision-making. Within HRC contexts, such real-time systems allow robotic teammates to dynamically adjust their behavior in response to detected CF levels, thereby promoting safer, more efficient, and human-aware collaboration.

To support the evaluation of CF during and after HRC, subjective assessment methods remain indispensable. Given the complex and dynamic nature of HRC, self-reported assessments provide valuable insights into users' internal experiences of CF, capturing subjective perceptions that may not be fully reflected in performance metrics

or physiological indicators alone [7, 8]. These assessments provide essential user-centered data, serving to validate and complement objective measurements collected via sensor data and system logs, thereby contributing to a holistic understanding of CF in HRC contexts.

For a comprehensive evaluation of CF in HRC, several well-established subjective assessment tools have been selected, each demonstrating robust psychometric properties and proven applicability in cognitively demanding environments. Table 1 summarizes the most widely used instruments, including NASA Task Load Index (NASA-TLX) [9], Fatigue Assessment Scale (FAS) [10], Mental Fatigue Scale (MFS) [11], and Visual Analogue Scale for Fatigue (VAS-F) [12]. These tools have been extensively applied across various domains and are increasingly utilized in HRC research [13, 14], where accurate assessment of users' CF is critical for ensuring safety, maintaining performance, and supporting effective collaboration.

As detailed in Table 1, each instrument offers distinct advantages and is suited to different phases of HRC, including post-task evaluations, in-task monitoring, and longitudinal fatigue tracking. Although these methods have inherent limitations, such as reliance on self-reporting and the potential to disrupt ongoing tasks, they remain essential components of HRC studies. Their ability to capture nuanced user perceptions of CF provides valuable insights not only for assessing cognitive states but also for informing the design of fatigue-aware, adaptive robotic behaviors aimed at fostering resilient, safe, and effective human-robot teams.

CF is a state of mental exhaustion characterized by diminished alertness and impaired cognitive function, posing significant challenges in high-demand domains such as HRC, where human performance, decision-making, and safety are critical. To address these challenges, researchers have increasingly adopted sensor-based methods for CF detection, leveraging physiological signals, behavioral cues, and speech patterns. Physiological measures, including EEG, ECG, and Electrodermal Activity (EDA), provide objective, online indicators of a user's cognitive state [17]. Among these, EEG is particularly sensitive to CF, and is widely recognized as one of the most accurate modalities for detecting brain-state changes associated with fatigue [18]. In parallel, heart rate variability derived from ECG and skin conductance captured via EDA or GSR reflect autonomic nervous system responses closely linked to CWL and CF [19].

In addition to physiological signals, behavioral monitoring approaches, such as vision-based tracking of gaze behavior, head pose estimation, or blink frequency, offer observable indicators of CF, although these methods are often influenced by environmental conditions and task configurations [20]. More recently, speech analysis has emerged as a promising, non-intrusive modality for CF detection, as CF affects prosodic and acoustic features, leading to slower speech rates and changes in voice quality [14]. Collectively, these multimodal sensing techniques enhance the capability to detect CF in HRC contexts, supporting the development of fatigue-aware robotic systems capable of dynamically adjusting task allocation, interaction modalities, or support strategies to maintain safe and effective collaboration [6].

In this chapter, we outline our methodology for selecting relevant research literature, focusing exclusively on studies where CF detection has been applied directly in HRC settings, either during data collection or within interactive scenarios. This

Table 1: Summary of widely used subjective cognitive fatigue (CF) assessment tools and their relevance to human-robot collaboration (HRC) contexts. Each instrument offers unique strengths suited to different assessment phases, including post-task evaluations, in-task monitoring, and longitudinal tracking. Despite inherent limitations such as reliance on self-reporting and the risk of interaction disruption, these tools remain integral to CF assessment in HRC studies, providing essential user-centered data that support the development of fatigue-aware, adaptive HRC systems.

Tool	Type	Key Features	Strengths in HRC Context	Limitations in HRC Context
NASA-TLX	Workload Assessment	6 dimensions (Mental, Physical, Temporal Demand, etc.) [9]	Assesses perceived workload during robot interactions; widely validated in HRC studies [13]	Not specific to CF; lacks sensitivity to accumulated fatigue
FAS	General Fatigue Scale	10-item unidimensional scale (physical + cognitive fatigue) [10]	Simple; useful for tracking general fatigue trends over work periods [15]	Less sensitive to immediate or task-specific CF during dynamic HRC tasks
MFS	Mental Fatigue Scale	15 items targeting cognitive, emotional, sensory fatigue symptoms [11]	Captures nuanced mental fatigue symptoms; suitable for post-task or longitudinal assessments in HRC [16]	Lengthier to administer; may not be practical for rapid assessment during tasks
VAS-F	Momentary Fatigue Scale	Visual analogue format (single or multi-item assessing fatigue/energy) [12]	Fast, sensitive to momentary fatigue; ideal for use during HRC tasks for adaptive feedback [14]	Provides only subjective perception; requires frequent user input which may disrupt flow

targeted approach ensures that the reviewed works address the unique demands and challenges of HRC environments, where safety, adaptability, and collaboration effectiveness are paramount. Furthermore, we present the rationale for prioritizing physiological and speech-based modalities over vision- or behavior-based approaches. While vision-based methods (e.g., gaze, posture, blink rate) and behavioral analyses (e.g., task performance metrics) have been explored for CF detection, these approaches are often highly sensitive to environmental variability, occlusions, and workspace configurations that may vary in HRC settings. In contrast, physiological signals and speech modalities offer more robust, unobtrusive, and context-resilient alternatives, providing reliable indicators of CF that can be seamlessly integrated into HRC systems to enable adaptive and fatigue-aware task management.

2 Offline Cognitive Fatigue Assessment in HRC

Offline CF assessment in HRC relies primarily on post-task evaluations, including subjective self-reports and performance analyses, to capture users’ perceived CF levels. These approaches offer valuable insights into the human teammate’s internal state, complementing physiological and behavioral data streams. Although inherently retrospective and limited in supporting immediate adaptation, offline assessments remain critical for validating machine learning (ML)-based models and establishing robust ground truth CF labels. [6]

Table 2: Summary of studies applying CF detection in HRC settings, sensor modalities, adaptability, and modeling approaches.

Article (Authors, Year)	Sensors/Modalities	Adaptability in HRC	Model Type (Classification/Regression)
Chand <i>et al.</i> (2024) [21]	Vision (camera, posture, gaze)	Yes	Classification (in digital twin framework)
Shah <i>et al.</i> (2023) [22]	Assumed physiological	Yes	Classification (with adaptation via HMM + RL)
Yao <i>et al.</i> (2024) [23]	EEG, video features (such as eye blinking and head posture)	Proposed	Classification (fatigue” and ”normal”)
Kalatzis <i>et al.</i> (2022) [24]	ECG	No	Classification
Zand <i>et al.</i> (2024) [25] and (2025) [6]	ECG, EDA, speech, behavior logs	No	Regression (Continuous CF estimation) ML
Dissanayake <i>et al.</i> (2022) [26]	EEG	No	None (signal analysis only)
Wan & Zhou (2023) [27]	Simulation-based (SAFTE fatigue model)	Yes (planning stage)	None (theoretical prediction model)

Table 2 provides an overview of studies applying CF detection in HRC contexts, highlighting the diversity of sensor modalities, the extent of adaptability integration, and the modeling approaches employed. [21–23, 27] will be discussed in section 3, since their research is based on adaptability in HRC. The majority of existing works have focused on classification models, typically distinguishing between discrete fatigue states, such as fatigued and non-fatigued. Regression-based approaches for continuous CF estimation remain rare, exemplified only by Zand *et al.* [25]. Physiological signals such as ECG, EDA, and EEG continue to dominate the modality landscape, while vision-based and behavioral approaches have been explored in concept proposals [21, 28, 29].

To the best of the authors’ knowledge, Zand *et al.* [6, 25] represents the first effort to develop and validate multimodal regression models for offline CF estimation in HRC, addressing a critical gap by enabling fine-grained CF estimation essential for adaptive and fatigue-aware HRC systems. Previous CF detection research in HRC has exclusively focused on classification models, categorizing CF into discrete states (e.g., fatigued vs. non-fatigued), thus limiting the potential for nuanced system adaptation.

Despite the indispensable role of offline assessment in model validation and system benchmarking, its retrospective and user-dependent nature limits its effectiveness for real-time adaptation. Assessments such as VAS-F and NASA-TLX [9] require active user engagement, potentially introducing task interruptions and delays in CF detection. Nevertheless, offline assessments remain an integral component of comprehensive CF evaluation frameworks, supporting the triangulation of subjective, physiological, and behavioral data while informing user-centered system design, interaction strategies, and acceptance studies [13].

The following sections, 2.1 and 2.1.2, explore our research on detecting CF through ECG, EDA, and speech sentiments using regression models within ML.

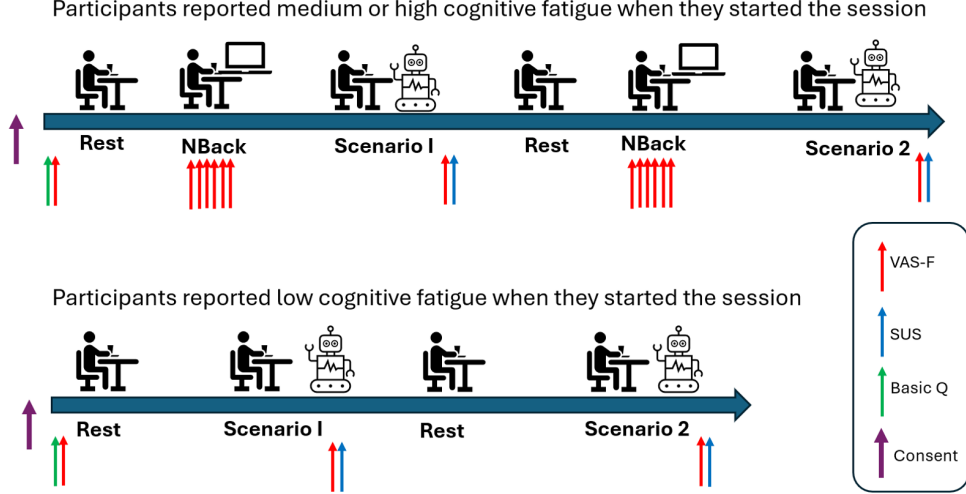


Fig. 1: Each visit starts with a consent form, VAS-F questionnaire, and a basic questionnaire. If the visit focused on medium or high fatigue, the top protocol was followed. If the visit focused on low fatigue, the bottom protocol was followed. It is worth noting that no participant reported a high CF level (above 70). Zand *et al.* [6]

2.1 A User Study on Cognitive Fatigue Assessment During Human-Robot Interaction

In the context of physiological signals, Cos *et al.* [30] demonstrated that both ECG and EDA are effective for detecting CF when integrated with ML models, achieving high classification accuracy. These modalities, as the authors point out, exhibit robustness across various contexts, which is especially beneficial in dynamic environments like HRC, where vision-based methods may face limitations such as occlusions and environmental variability. Additionally, Heaton *et al.* [31] highlighted that speech features, such as articulator velocity coordination and vocal creak, combined with arousal measures like electrodermal activity, can be predictive of performance changes linked to CF. These findings reinforce the potential of these physiological and speech-based modalities as reliable, unobtrusive tools for real-time CF assessment, crucial for enabling adaptive, fatigue-aware task management in HRC systems.

In this study, the HRC system, termed RoboCogniSense, facilitated collaborative manipulation tasks based on the reported CF level, combined with N-back cognitive tasks [32], which were designed to enhance CWL and induce CF. Sixteen participants were involved, with ages distributed as follows: 4 participants aged 18 to 30, 5 aged 31 to 40, 3 aged 41 to 50, 2 aged 51 to 60, and 2 aged over 60 years. The group consisted of 9 females, 5 males, and 2 non-binary individuals. Participants were recruited from non-STEM fields at Santa Clara University, and none of them were familiar with robots, in order to minimize bias. Each participant interacted with RoboCogniSense during three visits, each lasting up to two hours, which included two sessions: meal



Fig. 2: Experiment setup for Human-Robot Collaborative Cooking

preparation and getting ready for work. Each visit began with participants signing the consent form and reporting their CF level using the VAS-F assessment. If their CF was below 35, it was considered low; if above 70, it was considered high; and if between 35 and 70, it was classified as moderate CF. None of the participants reported their initial CF as being high.

Figure 1 illustrates the system protocol. The top protocol represents the case when participants report a medium or high CF level, while the bottom one corresponds to reporting CF as low. The key difference between the two diagrams is the inclusion of the N-back task in the top protocol, with the rest being identical. After completing the basic questionnaire and reporting their initial VAS-F level, participants rest for 5 minutes. For those with high or medium initial VAS-F levels, they perform 6 sets of the N-back task to induce CF, and after each set, they report their CF level using the VAS-F assessment. Participants then interact with the robot in two scenarios. To minimize bias, the order of the scenarios is alternated between participants. To ensure consistency across the study, participants follow a scripted speech format, such as: “Give me the cheese” or “Give me the car keys.” Each command was given by tapping the HyperX QuadCast [?] microphone

Offline CF assessment was conducted using VAS-F, administered before and after the pre- and post-interaction sessions, and after each N-back block. This approach enabled the monitoring of participants’ perceived CF evolution throughout the study phases, providing sensitive and easy-to-administer subjective data that served as the primary ground truth for developing and validating CF detection models. To support the multimodal modeling approach, continuous physiological and speech data were collected in parallel during the collaborative sessions. Physiological signals were included ECG to capture cardiac activity and EDA to assess sympathetic arousal, while speech data were recorded to extract prosodic and acoustic features indicative of

CF. These multimodal data streams were synchronized and stored using Robot Operating System (ROS) infrastructure, enabling precise timestamping and data alignment for offline analysis.

2.1.1 Cognitive Fatigue Assessment using Physiological Wearable Sensors

Building on this dataset, regression models were developed to estimate continuous CF levels using the self-reported VAS-F scores as the target variable. Several ML algorithms, including random forest regression, support vector regression, and gradient boosting, were evaluated. The best-performing model achieved an Root Mean Square Error (RMSE) of 4.85 on the normalized VAS-F scale and an R-squared (R^2) of 0.89, demonstrating the feasibility of highly accurate continuous CF estimation from multimodal data. Notably, the feature importance analysis revealed that speech prosodic features and ECG-derived metrics contributed most significantly to the model’s predictive power, while EDA offered complementary information in detecting CF fluctuations.

2.1.2 Cognitive Fatigue Assessment using Speech Signals

It is also interesting to investigate whether the speech signal can provide a modality for detecting CF. While there is research focused on identifying emotions through speech in Human-Robot Interaction (HRI) [33–35], the authors were unable to find related work on assessing CF during HRI through speech signals. For the CF assessment using the speech signal, the first step is to extract temporal and spectral features that are shown to vary systematically under fatigue, as follows:

- The speech ratio, the articulation rate, and the phonation time are selected as temporal characteristics based on the findings of Dias et al. [36], in which spontaneous speech in individuals with multiple sclerosis was analyzed. The authors reported that a lower speech ratio and a slower articulation rate were significantly associated with fatigue states. Phonation time, defined as the cumulative duration of voiced speech, is also found to decline with increasing fatigue, reflecting longer pauses and reduced verbal output.
- Formant frequencies (F1–F3) are selected due to their ability to capture articulatory changes related to fatigue-induced muscular control changes. Studies such as [37] and [38] have demonstrated that formant trajectories, particularly during sustained phonemes, show measurable deviations under fatigue. These frequencies represent the resonant characteristics of the vocal tract and are sensitive to changes in articulation clarity and vowel centralization.
- The fundamental frequency (F0), or pitch, is extracted as it correlates with vocal effort and neuromuscular control. Dias et al. [36] reported that F0 variability decreased with fatigue, consistent with a flatter prosodic contour and reduced vocal energy. This makes F0 a strong candidate for modeling fatigue-related vocal monotony.

- Jitter and shimmer, which quantify short-term variations in pitch and amplitude, respectively, are incorporated as measures of voice stability. These perturbation-based features have been associated with reduced fine motor control of the larynx under fatigue. The same MS-fatigue study found both jitter and shimmer to be among the top predictors of cognitive fatigue based on voice samples [36].
- Harmonics-to-noise ratio (HNR) is selected as a proxy for vocal clarity and breath control. A lower HNR indicates increased noise energy in the signal and is commonly observed in fatigued speech, where breathiness and vocal strain become more pronounced. Prior work in [36] used HNR to successfully distinguish between high- and low-fatigue conditions.
- Mel-Frequency Cepstral Coefficients (MFCCs) are chosen due to their widespread use and effectiveness in modeling the short-term spectral envelope of speech. Multiple studies [37, 39, 40] have demonstrated significant changes in MFCC vectors as fatigue increases, with intra-subject correlation of MFCC patterns decreasing sharply across fatigue levels. Their inclusion enables data-driven modeling of subtle spectral shifts not easily captured by prosodic or formant-based features alone.
- Gender is proven to affect the tone, frequency, and type of features (CITE). Therefore, any features developed from the audio data would be different between genders and thus affect classification accuracy if not taken into account. During the data collection portion of the study, the gender of each participant was recorded (Male, Female, Non-Binary). To correctly input gender as a feature, one hot encoding is used. If gender were instead represented numerically as Male = 0, Female = 1, Non-Binary = 2, this would cause any ML model to assume an ordering and distance between the numbers. This order is arbitrary, 2 is not greater than 1, and so forth. Instead, one-hot encoding creates a binary feature column for each gender. The model sees only “is Female?”, “is Male?”, “is Non-binary?”. There is no implied order or distance.

As we are interested in detecting CF, it is important to have a target value. As our speech commands are short (i.e., X seconds), classification may be more appropriate based on the number of datapoints available. Therefore, the target variable is if the CF level is Low Frequency (LF) or **MF!** (**MF!**). This was easily done as during the data collection process each participant’s individual visit information was already classified as either medium or low fatigue. To reiterate, MF is considered a VAS-F score above 30, while LF is any score equal to or below. As this becomes a binary classification problem, MF is considered a '1' while LF is a '0'.

Table 3: Summary of Speech Features Initially Adopted

Feature	Description / Rationale
Speech Ratio	Proportion of total speaking time to total recording time, quantifying the amount of pausing relative to voiced speech.
Articulation Rate	Speed at which speech sounds are produced (syllables per second); slows with increasing cognitive fatigue.
Phonation Time	Cumulative duration of voiced segments; shorter phonation and more/longer pauses indicate fatigue-related changes in vocal effort.
Formant Frequencies	Resonant frequencies of the vocal tract (typically F1–F3). Shifts or reduced distinctness can reflect articulatory changes due to fatigue.
Fundamental Frequency (F0)	Pitch of the voice. Fatigue often reduces F0 variability and can lower mean pitch.
Jitter and Shimmer	Micro-perturbation measures of pitch (jitter) and amplitude (shimmer) stability; higher values denote diminished vocal control.
Harmonics-to-Noise Ratio (HNR)	Ratio of periodic (harmonic) energy to aperiodic noise energy in voiced speech; lower HNR suggests breathiness or vocal strain.
Mel-Frequency Cepstral Coefficients (MFCCs)	Compact spectral representation derived from the short-term power spectrum via Mel-scale filter banks, log compression, and DCT; captures vocal-tract shape and spectral changes sensitive to fatigue.

The development of cognitive-aware HRI systems for Activities of Daily Living (ADL) requires accurately detecting and predicting CF to ensure safety and effectiveness. CF can impair task performance and increase the risk of errors or injuries during interactions. Previous research has utilized dermally applied ECG and Electrodermal Activity (EDA) data combined with N-Back tests and subjective Visual

Analogue Scale for Fatigue (VAS-F) assessments to model CF. However, challenges remain in achieving accurate CF detection through less invasive and more practical methods. This research aims to use speech and audio data to train classification Machine Learning (ML) models to advance CF detection in HRI systems. To the best of our knowledge, this will be the first attempt to recognize CF from speech data.

As mentioned this research will use existing ML classification models to detect either low CF (LF) or medium CF (MF). This threshold between LF and MF was set by previous researchers on the scores of a participant’s Vas-f test. Any Vas-f score above 30 is deemed MF and any equal or below is LF. Hence, by creating a binary classification as opposed to continuous target variable, this significantly increases the potential accuracy of any ML applications. Future research will be implementing a regression models for modeling CF. However due to the time constraints of this research, the first implementation will just focus on binary classification as a proof of concept that CF is possible from speech. Furthermore, the user ID and previous CF levels will not be used as input features. The goal is to predict CF from ‘blind’ speech: any speech segment regardless of user or context. Therefore, the feature extraction and selection process is paramount in order to distinguish CF from speech.

2.1.3 Initial Literature Review

The first step to understand what features to extract from speech was to conduct a literature review on what similar progress has been made on regarding feature extraction from speech. The following selected temporal and spectral features were shown to vary systematically under cognitive or physical fatigue. Table 4 lists and briefly explains the initial features chosen.

The speech ratio, the articulation rate, and the phonation time were included as temporal characteristics based on the findings of [36], in which spontaneous speech in individuals with multiple sclerosis was analyzed. The authors reported that a lower speech ratio and a slower articulation rate were significantly associated with fatigue states. Phonation time, defined as the cumulative duration of voiced speech, was also found to decline with increasing fatigue, reflecting longer pauses and reduced verbal output.

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the top predictors of cognitive fatigue based on voice samples [36]. Harmonics-to-noise ratio (HNR) was selected as a proxy for vocal clarity and breath control. A lower HNR indicates increased noise energy in the signal and is commonly observed in fatigued speech, where breathiness and vocal strain become more pronounced. Prior work in [36] used HNR to successfully distinguish between high- and low-fatigue conditions.

Finally, Mel-Frequency Cepstral Coefficients (MFCCs) were chosen due to their widespread use and effectiveness in modeling the short-term spectral envelope of speech. Multiple studies [37, 39, 40] have demonstrated significant changes in MFCC vectors as fatigue increases, with intra-subject correlation of MFCC patterns decreasing sharply across fatigue levels. Their inclusion enables data-driven modeling of subtle spectral shifts not easily captured by prosodic or formant-based features alone.

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2.1.4 Speech Isolation and Audio Trimming Process

The first main issue encountered during data preprocessing is that of microphone taps being included in the recorded audio files. In the study the user, physically taps the head of the microphone to unmute and mute themselves. The sensitivity of the microphone causes this tap by the user to be recorded in the audio recordings. This

is seen as a high frequency spike either preceding the spoken phrase or after. It is important to note that not all audio recordings from the data collection process have these spikes. However, the majority of the over 400 audio recordings do have either an initial spike, final spike, or both. In addition to this, the audio recordings are not all the same length. Although each phrase only takes a few seconds at most to say, the actual recorded audio files can range from a few seconds to 3-4 minutes. These longer files, however, do not contain any additional audio to the spoken phrase and potential microphone tap. The additional length is empty without any audio. Again, in a similar fashion to the microphone tap, this issue does not occur for every file and the length of the added empty audio is not consistent. This issue seemed to have arisen in the data collection process from a difference in when the audio is recorded and when the user taps the microphone. Figure 3 exemplifies the audio waveform of both issues. As this study focuses on detecting cognitive fatigue from spoken speech, any additional noise or redundant audio will negatively affect any machine learning model’s performance. Thus, a method for isolating the spoken phrase was needed. It had to be dynamic and flexible enough to remove microphone taps, empty sections, and noise while still refraining from cutting files that do not contain any issues. Thus the following pipeline was developed:

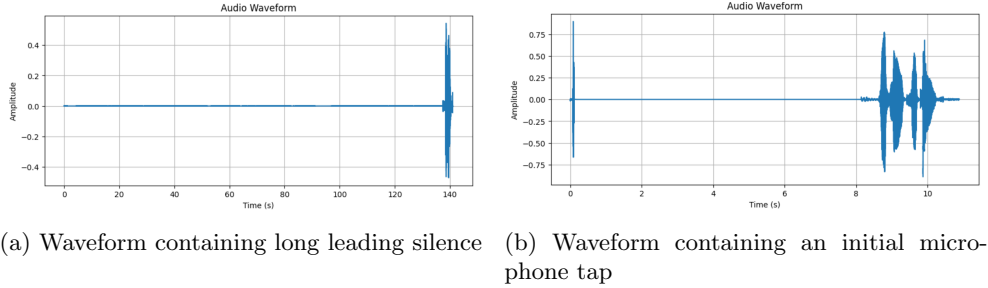


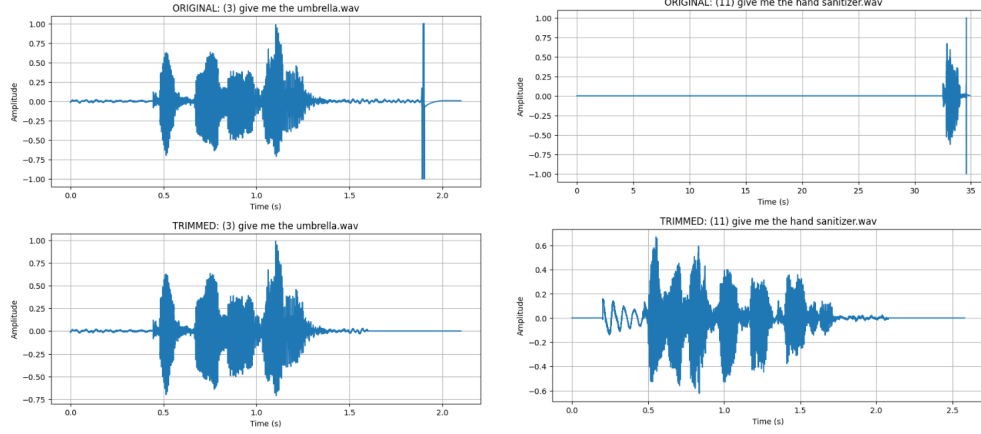
Fig. 3: Raw recordings before trimming: (a) excessive leading silence; (b) high-amplitude microphone tap.

1. **Load & book-keeping** – Import and read the raw waveform at a sampling frequency of 16 kHz with `librosa.load` and record the raw duration.
2. **Peak-normalize for Voice Activity Detection** – Librosa converts the waveform’s amplitude to be on the interval $[-1,1]$. However, in order to use WebRTC’s Voice Activity Detection (VAD) library the waveform must be transformed from floating-point numbers into a 16-bit, full-scale integer format. In 16-bit signed pulse code modulation (PCM) every sample is stored as an integer that can range from $\min = -2^{15} = -32768$ to $\max = 2^{15} - 1 = 32767$. Hence the signal is scaled via:
$$x_{\text{PCM}} = \frac{x}{\max|x|} \times 32767.$$
3. **Voice-activity detection** – The signal is chunked into 20 ms windows (frame size = 320 samples @16 kHz). The WebRTC-VAD is set to the ‘balanced mode

2" (balanced false-accept/false-reject). The VAD classifies each window as speech. The result is a Boolean vector `voiced[i]`, a voiced/unvoiced mask.

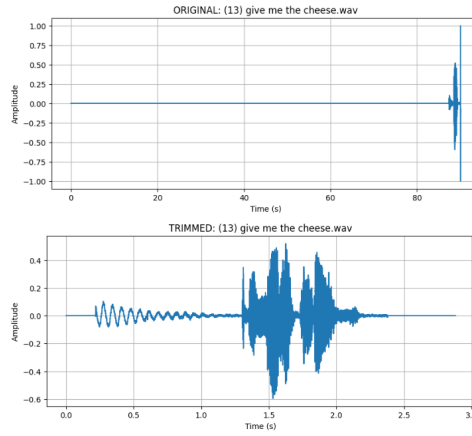
4. **Region-of-interest cut** – Using the `voiced[]` array, the block of continuous speech (the spoken phrase) is isolated by locating the first and last True in a continuous block of frames. A 200 ms pad is symmetrically added to first and last 'speech' index so that onset and offset consonants are not clipped.
5. **Quality guards** - Checks if the resulting audio segment contains only actual speech or any additional random noise.
 - If the remaining speech is shorter than 250 ms it is probably just a click, so the function gives up and returns an empty array.
 - The the last half-second of the audio excerpt is examined. If a spike is present which is more than four times louder than the rest of the excerpt, that part is set to 0. This accounts for any residual taps when the user mutes their microphone.
 - The last 30 ms are gently faded to zero so their is not a hard edge which could produce clicks or distort energy-based features during feature extraction.
6. **Persist & audit** – The cleaned audio is written to the disk. It resulting waveform is also plotted in comparison to its original untrimmed audio file. Additionally, both the original and trimmed durations so later stages can skip empty results.

The testing of the pipeline was conducted on the 3rd visit of participant ID6. This sample set was chosen as the majority of the files contained excessive silence and/or microphone taps. Figure 4 shows a sample of the resulting waveforms from the trimming process. Notice how in Figure 4 (a) the final mic tap is 'zeroed out' and removed. Figure 4 (b) and (c) demonstrate both successful audio trimming and tap removal.



(a) File 3 – original vs. trimmed

(b) File 11 – original vs. trimmed



(c) File 13 – original vs. trimmed

Fig. 4: Three representative waveforms before (top trace) and after automatic trimming (bottom trace).

2.1.5 Feature Extraction and Reduction

The first step in feature extraction is to extract the target variable: CF level. This was easily done as during the data collection process each participants individual visit information was already classified as either medium or low fatigue. To reiterate, MF is considered a VAS-F score above 30 while LF is any score equal or below. As this becomes a binary classification problem, MF is considered a '1' while LF is a '0'.

The next feature unique to this study is the use of gender, Gender is proven to affect the tone, frequency, and type of features (CITE). Therefore, any features developed from the audio data would be different between genders and thus affect classification accuracy if not taken into account. During the data collection portion of

the study, the gender of each participant was recorded (Male, Female, Non-Binary). To correctly input gender as a feature, one hot encoding was used. If gender was instead represented numerically as Male = 0, Female = 1, Non-Binary = 2, this would cause any ML model to assume an ordering and distance between the numbers. This order is arbitrary, 2 is not greater and 1 and so forth. Instead, one hot encoding creates a binary feature column for each gender. The model sees only “is Female?”, “is Male?”, “is Non-binary?”. There is no implied order or distance. Figure 5 shows the resulting columns.

Gender_Female	Gender_Male	Gender_Non-binary
1	0	0
1	0	0
1	0	0
1	0	0
1	0	0

Fig. 5: One-hot-encoded gender variable

As mentioned in the initial literature review the following features were initially extracted. The individual audio file from each participants visit was first fed into the audio trimming pipeline. Subsequently, using the librosa and parselmouth python libraries the listed features were extracted from the trimmed audio. It is important to note that Mel Frequency Cepstral Coefficients (MFCCs) were not included in this initial feature set.

1. Intensity
2. Speech Ratio
3. Articulation Rate
4. Jitter
5. Shimmer
6. Harmonics-to-Noise Ratio (HNR)
7. Gender_Female (one-hot encoded)
8. Gender_Male (one-hot encoded)
9. Gender_Non-binary (one-hot encoded)
10. Fundamental Frequency Mean (F0 Mean)
11. Fundamental Frequency Standard Deviation (F0 std)
12. Formant 1 (F1)
13. Formant 2 (F2)
14. Formant 3 (F3)

Before using any ML classification models, a correlation matrix was created from the resulting feature set to ensure good model performance. A correlation matrix is a table showing correlation coefficients between pairs of variables. Each cell in the matrix shows the degree to which two variables are linearly related, using the Pearson correlation coefficient. This coefficient ranges from: +1 (Perfect positive correlation), 0 (No linear correlation), -1 (Perfect negative correlation). The goal of creating this matrix is to predict CF Level, so the focus is on how each feature correlates with CF Level. Figure 6 shows the results of this matrix.

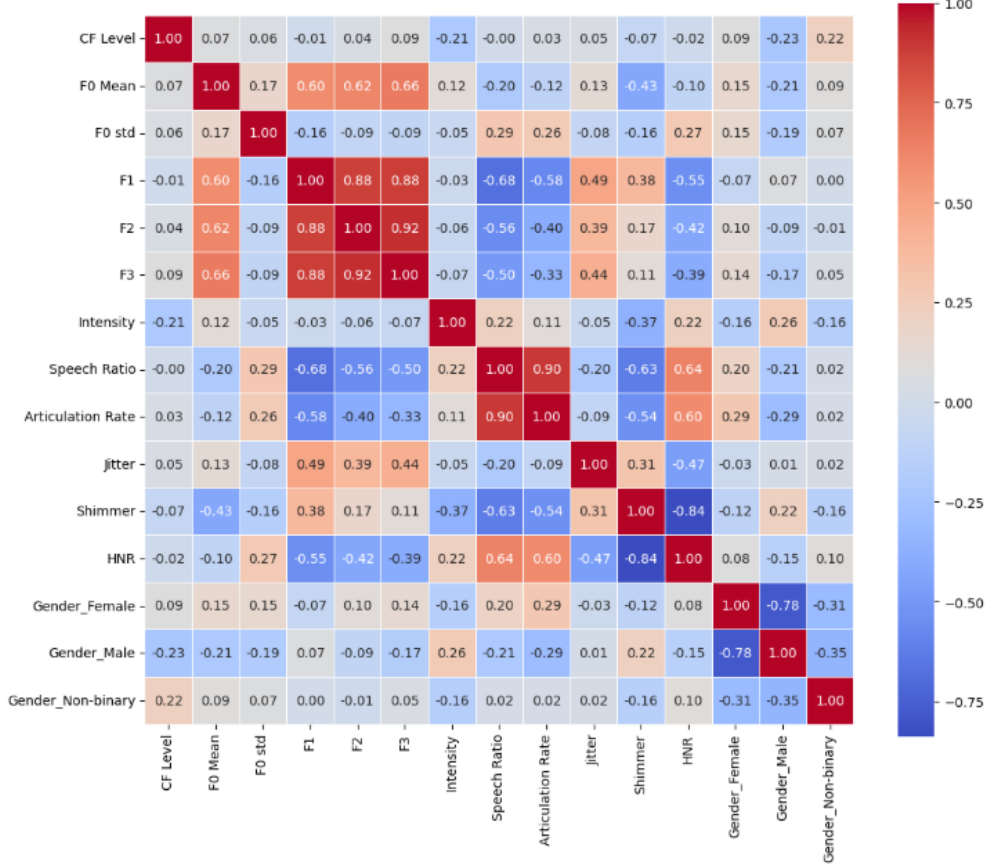


Fig. 6: Correlation Matrix of Initial Feature Set

From the Figure 6, F0 Mean, F1, F2, F3, and Articulation Rate have very low correlations (close to 0) with CF Level. These features likely do not provide meaningful signal about fatigue levels. Conversely, Shimmer and Intensity show moderate negative correlation with CF Level (Shimmer = -0.07, Intensity = -0.21), suggesting they may be slightly informative. Gender-Non-binary shows a positive correlation (= 0.22),

which is likely demographic-based rather than physiological. However, it still could help the model generalize better. Gender-Male shows a negative correlation (-0.23), reflecting the possible vocal differences across genders.

Therefore, based on these results, the Formants (F1, F2, F3) and pitch-based features (F0 mean/std) were dropped from the feature set despite their consistent use in speech based analysis. They showed a very low correlation with CF level in your dataset. Additionally, Articulation Rate and Speech Ratio were also dropped as they showed low correlation and were potentially redundant with jitter/shimmer/HNR. This left only jitter, shimmer, and the gender columns. These features were too poorly correlated with the target variable CF level to create a feature set. Therefore, additional features needed to be added. Thus Mel Frequency Cepstral Coefficients were added.

2.1.6 Mel Frequency Cepstral Coefficients

Mel Frequency Cepstral Coefficients (MFCCs) are consistently used in research relating to speech and machine learning. Even the initial short literature conducted proved that MFCCs are extremely common among current research. MFCCs are based on the Mel scale which is designed to match how humans perceive sound rather than how frequencies are physically distributed. Furthermore, human Hearing is non-linear. The human ear hears lower frequencies (e.g., 100-500 Hz) with greater sensitivity than higher ones. At higher frequencies (above 1000 Hz), our ability to distinguish frequency changes decreases. The Mel scale mimics human perception. Instead of using the linear Hz scale, the Mel scale spaces frequencies above 1000 Hz logarithmically. Frequencies below 1000 hz are spaced lineary ($1 \text{ hz} = 1 \text{ Mel}$). The Mel scale is commonly used in speech processing as converting to the Mel scale helps focus on important speech frequencies while reducing irrelevant high-frequency noise. This improves speech recognition, speaker identification, and audio classification tasks. MFCCs leverage this scale to extract features that capture the timbral aspects of speech. Specifically, MFCCs are features extracted from a signal that has been filtered through Mel-scaled frequency bands and further transformed using a logarithm and Discrete Cosine Transform (DCT). This makes MFCCs compact, decorrelated, and suitable for ML modeling. In this study, MFCCs were extracted using the Librosa library with the following pipeline:

- Audio is divided into overlapping frames of 50 milliseconds.
- Pre-emphasis of high frequencies
- Framing and windowing
- Fast Fourier Transform (FFT)
- Apply Mel filterbanks
- Take logarithm of the filtered energies
- Apply Discrete Cosine Transform (DCT)

This steps of this pipeline were adopted from (STATE STUDIES HERE) in the intial literature review. Furthermore, since each file still varied in length (even after trimming), any MFCCs had to be representative of the whole audio file. MFCCs computed on individual frames would lead to different amount of rows in the resulting

dataset. Each row would represent a distinct timestep in the file and this would transform our classification problem into a times series forecasting model (e.g. LSTM), which is not the goal of the study. As such a fixed-size representation was instead necessary. This was obtained by computing the mean and standard deviation of each MFCC across all frames in a file. This ensures consistency across samples for machine learning.

To address this, MFCCs were computed across all 50ms frames of each file using the Librosa library. Specifically the top 20 coefficients (nmfcc=20) were extracted per frame. These 20 coefficients correspond to the lowest 20 dimensions of the Discrete Cosine Transform (DCT) applied to the log Mel spectrum. Essentially these capture the most important parts of how we hear speech. Lower-order coefficients describe the overall shape of the sound. For example, this refers to how the participants vocal tract (mouth and throat) affects the resonant frequencies in their voice. Conversely, higher-order coefficients capture fine details (often noise). In practice, retaining just the first 12-20 MFCCs is sufficient for most speech analysis tasks, as higher coefficients tend to encode less useful or noisy information.

To convert the resulting MFCC time-series matrix into a consistent feature vector per file, 7 'summary' statistics were computed across the time axis (i.e., for each coefficient across all frames). Specifically, the mean, standard deviation, 25th percentile, median (50th percentile), 75th percentile, skewness, and kurtosis were calculated. These seven features capture both the central tendency and distributional shape of each MFCC coefficient over time. This results in a feature set of shape $7 \times 20 = 140$ per file (row in a dataframe).

Additionally, guard checks are included in the implementation to detect files that are too short or contain malformed MFCC arrays (e.g., empty, incorrect dimensionality). In such cases, zero arrays are returned to avoid throwing errors during future model training.

The resulting correlation matrix is seen in Figure 7. Figure 7 shows how CF Level has weak correlations with all features (most values were between -0.2 and 0.2). This suggests that no single feature has a strong linear relationship with cognitive fatigue. However, this could suggest that combinations of features may still be useful for classification. Furthermore, most of the MFCC features are strongly correlated with one another. Especially the adjacent MFCC mean features (e.g., 1-5) have correlate highly with each other (>0.9). Additionally, the MFCC std features are also clustered. This suggests that such high multicollinearity could lead to redundant features and that feature reduction could benefit ML model performance. Overall, no single feature dominates in predicting CF Level, but combinations (especially from MFCC means and stds) could hold value.

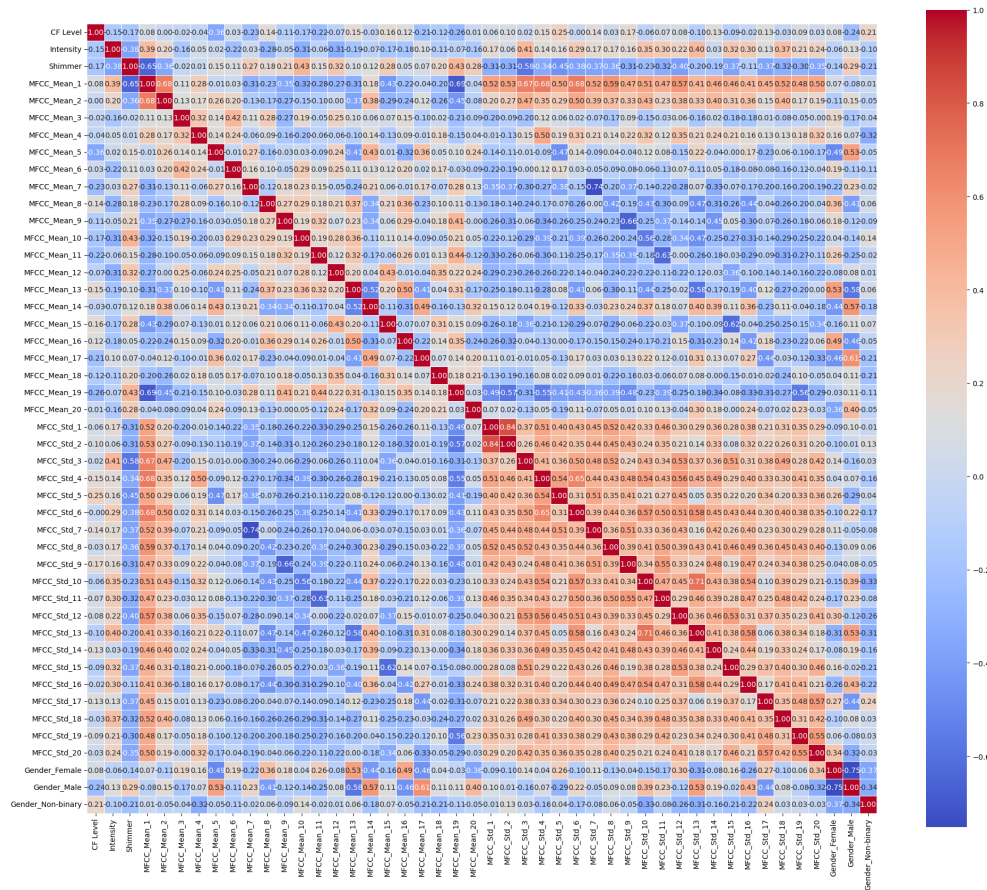


Fig. 7: Correlation Matrix of MFCC Feature Set

2.1.7 Dataset Splitting Procedure

In any ML related research or project, the feature set must be split into training and test sets. The test set acts as a benchmark and allows for model performance reports. As mentioned previously, one key aspect of this proposed research is to create models blind to which ever participant has spoken. Therefore, a randomized shuffle of the feature (sklearn train test split), would cause data leakage. The data from the same speaker would appear in both the training and test sets. Any performance metric (accuracy) would be artificially inflated as the model would have already 'seen' (been fit too) a given participants voice.

To prevent this, the feature set was first sorted by participant ID and visit number. Then, the number of samples corresponding to each ID was computed. A cumulative sum of these counts was calculated to determine how many samples would fall into the training versus the testing set. For the train test split an 80/20 split was defined. Iterating through the cumulative counts, each ID was assigned to either the training

or testing set. If the cumulative sum was less than or equal to 80% of the total data, the ID was assigned to the training set. Once the cumulative sum surpassed the 80% threshold, the remaining IDs were assigned to the testing set.

2.1.8 Binary Classification Using Machine Learning

Using the previously mentioned MFCC feature set, 4 supervised learning ML models from the Sklearn library were implemented and evaluated for binary classification of CF (LF or MF). The first model used is Support Vector Machine (SVM). SVM seeks the decision surface that maximises the margin between the two cognitive-fatigue classes. When an RBF kernel is used, the algorithm projects the 50-dimensional feature vectors (20 MFCC means, 20 MFCC standard deviations, shimmer, intensity and one-hot gender indicators) into a higher-order space in which a nearly linear separation often emerges. SVMs handle moderate sample sizes well and are comparatively immune to the curse of dimensionality, making them a natural first choice for a data set whose dimensionality is respectable but not extreme.

The second model used is Logistic Regression (LR). As this is a linear model it is the simplest and thus the baseline for performance. LR fits a hyperplane whose distances are input into the sigmoid function. This returns class probabilities (0 to 1). Because the model is linear in the parameters, every coefficient can be interpreted as the log-odds change associated with a one-unit rise in a given feature, holding the others fixed. Including LR therefore serves two purposes: (i) it establishes how much predictive power can be obtained without invoking non-linear machinery, and (ii) it furnishes a transparent yardstick against which more sophisticated methods may be judged.

The third model is Random Forest Classifier (RF). The RF model aggregates a large number of decorrelated decision trees. Each tree is 'grown' on a sample and random subset of features. RF can capture intricate feature interactions without extensive hyper-parameter tuning while remaining resistant to over-fitting.

The final model is Extreme Gradient Boosting (XGBoost). XGBoost incrementally creates a group of shallow trees. Each tree is trained to correct the residual errors of the preceding group. Compared with bagging-based approaches, boosting often delivers higher predictive accuracy on structured, mixed-scale data such as this.

Each model was implemented by using the Sklearn library. However, each pre-made model still requires hyper-parameter tuning in addition to training on the training set. As such as 3-fold cross-validation strategy, similar to a grid search, was used to search for the optimal model parameters for this feature set. Specifically, 3 folds strikes a balance for the size of this data set (around 360 recordings). Using fewer folds would reduce the reliability of the validation scores, whereas 5-10 folds would be similarly be inaccurate. It would leave each training subset so small that the variance of the estimates would grow and the search would become too slow. For the SVM, the regularization constant C (0.1, 1, 10), the kernel type (linear vs. RBF) and the γ parameter for the RBF kernel (set to scale, i.e. $1 / \text{number-of-features}$) were tuned. For the RF, the number of trees ($n_{\text{estimators}} = 50, 100, 200$), the maximum tree depth ($\text{max depth} = \text{None}, 10, 20$) and the minimum number of samples to split an internal node ($\text{min samples split} = 2, 5, 10$) were tested. As Logistic Regression is

the simplest model uses the sigmoid function, no model parameters are needed. For XGBoost, number of boosting rounds (n estimators = 50, 100, 200), the maximum depth of each tree (max depth = 3, 5, 7) and the learning rate (learning rate = 0.01, 0.1, 0.2) were tested.

For every model the grid point that produced the highest mean CV accuracy was automatically selected. Subsequently, each model was then re-trained on the full training set using the corresponding optimal hyper-parameters. Finally, classification reports were produced by comparing the trained models to the test set. The results are seen in Table ??.

2.1.9 Evaluation Metrics

The following metrics were used for evaluating the performance of the classification models. For reference, the positive class refers to the HF or '1' class in the output. The negative class refers to LF or '0' class in the output. Thus, for example, true positive refers to an instance when the model correctly predicts HF (1).

2.1.10 Precision

Definition: Precision is the ratio of true positive predictions to the the total number of positive predictions. Precision evaluates how well a model avoids misclassifying negatives instance as positive. It focuses on the quality of the positive predictions. For example, out of all positive predictions, precision looks at how many were actually positive.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (1)$$

2.1.11 Recall

Definition: Recall, or also known as Sensitivity, is the ratio of true positives to the actual positives (true positives and false negatives). Recall evaluates how many positive instances were correctly predicted. Recall focuses on the quantity of of actual positives. For example, out of all actual positives, recall focuses on how many were correctly predicted.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (2)$$

2.1.12 F1 Score

Definition: The F1 score combines precision and recall. It considers both false positives and false negatives. It is the harmonic mean of precision and recall. It is useful when you have an uneven class distribution.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

2.1.13 Support

Definition: Support refers to the number of actual occurrences of each class in the dataset.

2.1.14 Modeling Results

Table 5 reports the confusion matrix and derived metrics for the SVM model. Of the 57 low-fatigue (LF) samples, the classifier correctly identified 27 as LF (true negatives) while mislabelling the remaining 30 as moderate fatigue (MF). Conversely, of the 25 MF recordings, 19 were correctly detected (true positives) and 6 were missed. These counts translate to a precision of 0.82 for LF—indicating that when the SVM predicts “low fatigue,” it is usually correct—yet only 0.39 precision for MF, showing that most MF predictions are in fact false alarms. Recall tells the complementary story: LF recall falls to 0.47, meaning more than half of the truly low-fatigue clips are erroneously escalated to MF, whereas MF recall rises to 0.76, so genuine fatigue is seldom overlooked. The unbalanced recall values produce a macro-averaged F1 of 0.56 and a global accuracy of 56%. In operational terms, this model behaves like a highly sensitive but poorly specific detector: it catches most fatigued speech but inundates the system with false-positive fatigue warnings, while simultaneously allowing many low-fatigue instances to be misclassified.

The Random-Forest ensemble in Table 6 demonstrates a more even error profile. Out of the 57 LF samples, 36 were correctly classified, yielding LF precision 0.77 and LF recall 0.63. For the MF class, 14 of 25 samples were captured, giving MF precision 0.40 and MF recall 0.56. Although MF precision remains modest, the forest recovers a considerably larger share of LF recordings than the SVM while maintaining comparable MF recall. Consequently, the weighted F1 climbs to 0.62 and overall accuracy to 61%. The improvement stems from the forest’s ability to exploit non-linear interactions among the MFCC means, standard deviations, shimmer, and intensity without over-fitting to the minority class. In practice the Random Forest reduces missed low-fatigue cases and cuts the flood of false MF flags compared with the SVM, providing a more balanced—if still imperfect—classification.

XGBoost’s results (Table 7) sit between the two previous models. The algorithm correctly labelled 34 of 57 LF clips (precision 0.76, recall 0.60) and 14 of 25 MF clips (precision 0.38, recall 0.56). Accuracy settles at 59%, and the macro-averaged F1 at 0.56 mirrors the SVM’s figure, though with a more symmetrical recall distribution (0.60 vs 0.56). The boosted trees evidently capture some nuanced feature patterns yet deliver no marked gain over the bagged forest, possibly because the data set is too small for the additional boosting capacity to manifest. The persistence of low MF precision indicates that class imbalance, rather than model bias, limits performance; approximately three out of five fatigue predictions remain false positives.

Across all three classifiers the dominant weakness is medium-fatigue precision: only 38–40% of segments flagged as MF are truly fatigued. Nevertheless, MF recall never drops below 0.56, so the system rarely overlooks genuine fatigue. The Random Forest offers the best compromise—highest overall accuracy, improved LF recall, and only a modest sacrifice in MF recall—making it the preferred choice for this study. The

SVM might be selected in scenarios where missing fatigue is deemed unacceptable and extra false positives can be tolerated, whereas XGBoost’s added complexity does not translate into superior performance given the present sample size and feature set.

Table 5: Classification Metrics of the SVM Model

	Precision	Recall	F1-Score	Support
LF	0.82	0.47	0.60	57
MF	0.39	0.76	0.51	25
Accuracy			0.56	82
Macro Avg	0.60	0.62	0.56	82
Weighted Avg	0.69	0.56	0.57	82

Table 6: Classification Metrics of the Random Forest Model

	Precision	Recall	F1-Score	Support
LF	0.77	0.63	0.69	57
MF	0.40	0.56	0.47	25
Accuracy			0.61	82
Macro Avg	0.58	0.60	0.58	82
Weighted Avg	0.65	0.61	0.62	82

Table 7: Classification Metrics of the XGBoost Model

	Precision	Recall	F1-Score	Support
LF	0.76	0.60	0.67	57
MF	0.38	0.56	0.45	25
Accuracy			0.59	82
Macro Avg	0.57	0.58	0.56	82
Weighted Avg	0.64	0.59	0.60	82

2.2 Select K Best

Due to the large number of features and multicollinearity present between them (stated in previous sections), a feature reduction method was deemed as an area for further exploration. Reducing the feature set would potentially reduce collinearity and allow the models to generalize better to unseen data. The selected feature reduction method is Sklearn's SelectKBest function. This function returns the top 'h' specified features that produce the best model performance. For SelectKBest a scoring function must also be specified. This study uses the ANOVA F-test for classification. ANOVA (Analysis of Variance) F-test is a statistical hypothesis-testing framework that can be applied either for regression or classification tasks. Due to the binary nature of the target variable, the classification version of ANOVA was applied. Essentially, ANOVA works by comparing between-group to within-group variance. With classification the groups are simply the class labels and the test asks whether the feature's mean differs across those classes. Moreover, SelectKBest uses ANOVA to evaluate each feature independently against the target, rank features by score, and keep the top k features.

Essentially, this pipeline works by first using the ANOVA test to measure how much each MFCC/intensity/shimmer variable's mean differs between LF and MF audio files. As stated the SelectKBest algorithm, retains the top k features. Furthermore, the feature ranking is recomputed from scratch on each training fold during cross-validation and is never exposed to the corresponding validation fold. This prevents any data leakage. A grid search then sweeps simultaneously over k (1 to the total number of features) and the hyper-parameters of three candidate classifiers—SVM (C, kernel type, γ), Random Forest (tree depth, number of trees, minimum split size) and XGBoost (learning rate, depth, estimators). For every grid point the entire pipeline is refit in five internal folds; mean macro-F1 across those folds determines the winning configuration. Finally, the best-scoring combination (e.g., Random Forest with $k = 20$ and 200 trees) is retrained on the train set and evaluated on the test set. Essentially, this process finds the combination of k and hyperparameters for each model that maximizes performance.

Following this grid search, each model converged on different hyper-parameters and feature counts. Firstly the SVM model converged to using a RBF kernel, a penalty C of 10, and feature set (k) of 44. Secondly, the Random-Forest settled on 200 trees, unrestricted depth, a slightly stricter node-split criterion (min samples split = 5), and far smaller feature set of $k = 20$. Finally, XGBoost preferred the same tree count (200) but compensated with a higher learning rate (0.10) and medium depth (5) while keeping $k = 23$ features.

Tables 8, 9, 10, show the classification report for the optimized feature set and models. Overall, each model's performance did not increase significantly in comparison to using the whole feature set. However, each model cracked an average accuracy of over 60%; roughly 2/3 accuracy in detecting CF levels.

Furthermore, the RF model delivered the highest macro-F1 (0.59) and top accuracy (0.62), thanks largely to balanced recall. The model accurately predicted 56% of the true MF clips while still identifying 65% of the LF clips. However, its MF precision (0.41) shows that many LF clips are still being mis-flagged as fatigued. The SVM model achieved the sharpest LF discrimination (precision 0.73, recall 0.70, F1 0.71)

but paid for that margin with the weakest MF performance (F1 0.38). This suggests the RBF hyper-surface is tilted toward the majority class. This could be the result of the k=44 feature set. The XGBoost model landed between the previous two models. Its gradient boosting predicted the same 56% of MF clips as the forest, but precision slipped to 0.39 and overall accuracy to 0.60. Further tuning is needed to correct the model’s bias toward the dominant LF label.

Table 8: Optimized SVM Model Results

	Precision	Recall	F1-Score	Support
LF	0.73	0.70	0.71	57
MF	0.37	0.40	0.38	25
Accuracy			0.61	82
Macro Avg	0.55	0.55	0.55	82
Weighted Avg	0.62	0.61	0.61	82

Table 9: Optimized Random Forest Model Results

	Precision	Recall	F1-Score	Support
LF	0.77	0.65	0.70	57
MF	0.41	0.56	0.47	25
Accuracy			0.62	82
Macro Avg	0.59	0.60	0.59	82
Weighted Avg	0.66	0.62	0.63	82

Table 10: Optimized XGBoost Model Results

	Precision	Recall	F1-Score	Support
LF	0.76	0.61	0.68	57
MF	0.39	0.56	0.46	25
Accuracy			0.60	82
Macro Avg	0.57	0.59	0.57	82
Weighted Avg	0.65	0.60	0.61	82

Overall, when compared to the previous results in Tables 5, 6, 7, the accuracy gains from SelectKbest and the grid search were marginal. The grid search pipeline added at most +1-5 points of accuracy. Additionally, the macro F1 score barely changed between the results. The nature of the small training set (around 330 audio clips) could have limited the optimization power of the hyperparameter tuning. In the optimized vs unoptimized models the MF recall stood at around 0.56. Thus, tuning cannot fix the shortage of MF examples. More data or class-balancing strategies would be required. Overall, hyper-parameter optimization only slightly improved the models but did change the optimal model. The Random Forest remained the best of the three (macro-F1 0.59), with SVM following, and XGBoost lagging slightly behind.

3 Online Cognitive Fatigue Assessment in HRC

Timely detection of CF is particularly critical for enabling adaptive HRC systems. In human teams, individuals often intuitively recognize when their teammates are fatigued and adjust their behavior accordingly, whether by redistributing tasks, offering assistance, or altering their communication styles. Similarly, in HRC, robots must be capable of perceiving and responding to the cognitive state of their human teammates in real-time. This allows them to adjust task allocation, interaction modalities, or pacing strategies to mitigate the effects of CF and maintain team performance, safety, and user trust [3].

As shown in Table 2, notably, only a few studies, such as those by Chand *et al.* [21] and Shah *et al.* [22], have incorporated CF detection into adaptive HRC frameworks, enabling robots to adjust their behavior based on the detected fatigue level. Several recent works, including those by Yao *et al.* [23] and Zhang *et al.* [41], have validated CF detection methods in laboratory settings without exploring adaptation or integration into real-time HRC systems. This highlights an ongoing gap in the field, where most studies remain focused on offline or observational analyses, with limited exploration of continuous CF estimation and fatigue-adaptive robotic behaviors in interactive, real-world HRC scenarios.

4 Open Challenges

Recap the open challenges

- Lack of Datasets with a diverse population - most datasets are small
- Generalization
- Real-time Processing
- Multimodal Fusion
- User Acceptance

5 Conclusion

This chapter presents

Acknowledgements. The material is based upon work supported by the National Science Foundation under Grant 2226165.

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